

JEE MAIN REVISION MASTER: MAGNETISM AND MATTER 2026 2 QUESTIONS 3.33% WEIGHTAGE

↑99.4%

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 18 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Kcet Magnetism And Matter - Core Application	18 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Kcet Magnetism And Matter - Core Application (Very High Yield — 18 PYQs)

Core Principle: Key problem pattern in Kcet Magnetism And Matter.

■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "Pick out the WRONG statements about magnetic substances. (χ = magnetic susceptibility) (μ_r = relative permeability). I. Substa..." [Ans: N/A]
- [2026 Shift] Q#2 (MCQ): "If a paramagnetic bar is brought near a bar magnet, then it is..." [Ans: N/A]
- [2025 Shift] Q#3 (MCQ): "Which of the following statements is true in respect of diamagnetic substances?..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

Pick out the WRONG statements about magnetic substances. (χ = magnetic susceptibility) (μ_r = relative permeability). I. Substances with $-1 \leq \chi < 0$ are diamagnetic. II. Substances with $\chi \ll 1$ are paramagnetic. III. Substances with $\chi \gg 1$ are ferromagnetic. IV. Substances with $\mu_r \gg 1$ are ferromagnetic.

- (A) I and II
- (B) III and IV
- (C) II and III
- (D) II and IV

Correct Answer: None

Solution Analysis: The wrong statements are II and III. For magnetic substances, remember these NCERT-based facts: Diamagnetic substances have small negative susceptibility: $-1 \leq \chi < 0$. So statement I is correct. Paramagnetic substances have small positive susceptibility: $0 < \chi \ll 1$. They do not have $\chi \gg 1$. So statement II is wrong. Ferromagnetic substances have very large positive susceptibility: $\chi \gg 1$. They do not have $\chi \ll 1$. So statement III is wrong. For ferromagnetic substances, relative permeability is also very large: $\mu_r \gg 1$. So statement IV is correct. Hence, the wrong statements are: II and III.